

203
MTH 302: Linear Algebra
Mid-examination 1: 03/09/2025,
Time: 12:00 to 13:00

Total points: 25

Problem 1. (2 points each)

- (a) State the Axiom of Completeness.
- (b) What is the Archimedean Property of the real numbers?

Problem 2. (4 points) Let $A = \{x \in \mathbb{R} \mid -1 \leq x \leq 1\}$. Let $R \subset A \times A$ be a relation from A to itself defined as follows:

$$\{(x, y) \mid x^2 + y^2 = 1\}.$$

Is R a function? You must justify your answer.

Problem 3. (5 points) Let $S \subset \mathbb{R}$ and let $s = \sup(S)$. Let t be an element of $\mathbb{R}$ such that $t \notin S$. Prove that $\sup(S \cup \{t\}) = \sup\{s, t\}$.

Problem 4. (6 points) Let A and B be subsets of $\mathbb{R}$. Let $a = \sup(A)$ and let $b = \sup(B)$. Let C be the subset defined as follows:

$$C = \{z \in \mathbb{R} \mid z = x + y \text{ for some } x \in A \text{ and } y \in B\}$$

Prove that C is bounded above and $\sup(C) = a + b$.

Problem 5. (6 points) Let $S = \{1/n + 1/m \mid m \in \mathbb{N} \text{ and } n \in \mathbb{N}\}$. Compute $\inf(S)$ and $\sup(S)$.